

Introduction

This project designing a circuit that could detect the sound of people's footsteps and output a voltage which can be used to light up a led in a corridor. When the light intensity is high enough during the day, the led are not activated regardless of whether there are people walking through the corridor. When the light intensity goes down, if someone walks by, the sound of footsteps will turn on the led.

Interface of blocks



First is passive bandpass filtering system. The system isolates our target frequency range(100Hz – 3000 Hz). The filter helps us to remove other sounds (very low frequency sound such as wind, distant noise; very high frequency noise which even inaudible). The output signal is then passes into an operational amplifier. This op-amp amplifies the voltage of the signal to a value sufficient to complete the subsequent circuits. The output of the ap-amp is then passes to a half-wave rectifier, which converts the signal into a DC signal. Diodes in the rectifier smoothing the input signal by ensuring that current flows in only one direction. Moreover, the rectifier has a delay effect. It delays the discharge time about 10 seconds, so that the led light does not flash because of frequent sound input, but continues to light up until pedestrians passed. The signal then enters the non-inverting voltage comparator, comparing with a reference voltage on the inverting input and outputs high level. Also, change the shape of the signal. Next the signal enters into the light control part, once the light intensity is high, Q1 will be active to ensure that the led will not turn on during the daytime. Finally, once all conditions are satisfied, the led works.

Circuit complete design

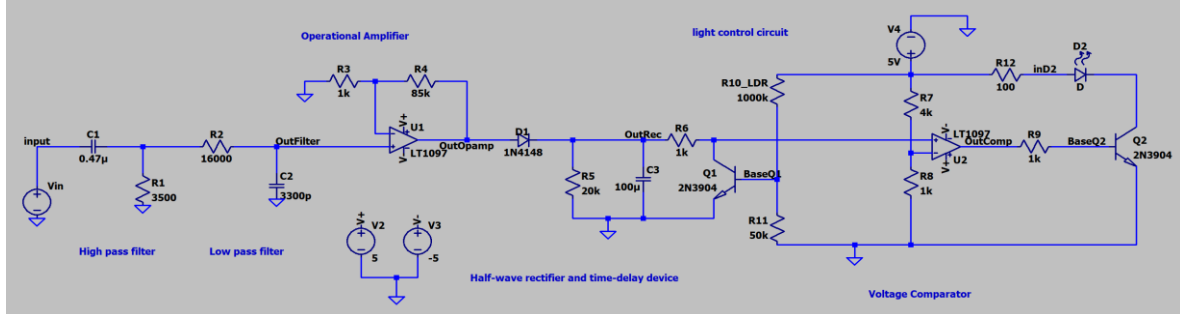


Figure a 1

Individual blocks

Input voltage(microphone)

According to Internet search and some electronic component sales platforms, the output voltage of a microphone ranges from a few millivolts to tens of millivolts. In this project, I assume that the input voltage is about 50 millivolts. This will determine how much magnification will be needed later to complete the circuit.

Bandpass filter

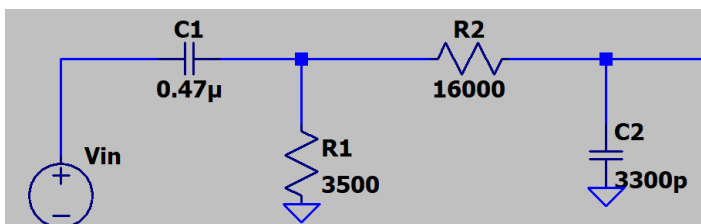


Figure b 1

For high pass filter part , $f_1 = \frac{1}{2\pi R_1 C_1} = 100 \text{ Hz}$, consider the most common available capacitor values[2], I set up the R_1 is 3500 Ohms, so the C_1 will be $0.4547\mu\text{F}$. The closest available capacitor of the same value is $0.47\mu\text{F}$, resulting in a cutoff frequency of 96.75 Hz.

For low pass filter part, $f_2 = \frac{1}{2\pi R_2 C_2} = 3000 \text{ Hz}$, setting up

the R_2 is 16k Ohms, so the C_2 will be 3315.73pF. The closest available capacitor value is 3300 pF, resulting in a cutoff

frequency of 3014.30 Hz. Thus, the resulting center frequency of the bandpass filter will be, $f_r = \sqrt{f_1 \cdot f_2} = 540 \text{ Hz}$.

The transfer function of the filter can be governed that assume $\tau = RC$, $H_1(s)$ is the transfer function of high pass filter and $H_2(s)$ is the transfer function of low pass filter. We can write: $H(s) = H_1(s) * H_2(s)$ so that $H(f) = \frac{j2\pi f\tau_1}{j2\pi f\tau_1 + 1} * \frac{1}{j2\pi f\tau_2 + 1}$.

Q factor is a dimensionless unit that measures the performance of a component or resonant loop. Simply the ratio of the ideal component to the loss present in the component. For this filter, it can be defined as: $Q = \frac{f_r}{f_2 - f_1} = 0.186$

Non-inverting Operational Amplifier

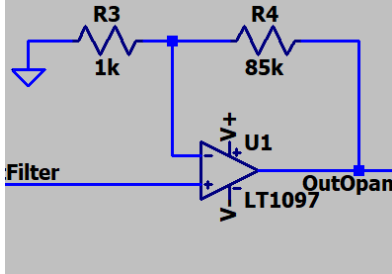


Figure b 2

Assume that V_1 is the feedback signal. We can calculate the voltage gain as follows:

$$V_1 = \frac{R_3}{R_3 + R_4} \times V_{out} \quad , \quad \text{and} \quad \text{Because of } V_1 = V_{in}, \text{ voltage gain } A_V = \frac{V_{out}}{V_{in}} \quad , \quad \text{the}$$

transpose to give $A_V = \frac{R_4}{R_3} + 1$. And we should also consider to avoid rippling. As the result, I chose R_3 equals to 1k ohms and R_4 equals to 85k ohms. The gain will be 86 and the signal will around 4 volts [2][3]

Half-wave rectifier and time-delay device

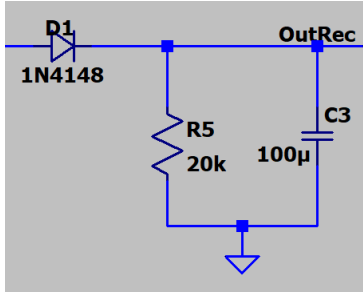


Figure b 3

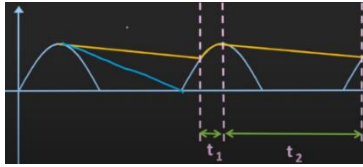


Figure b 4

Choice of the capacitor value needs to fulfil several requirements. Firstly, we can see the figure on left hand side, t_1 is the time when charge the capacitor, t_2 is the time when discharge the capacitor. To avoid the situation like the blue line in figure b 4, its time constant is much longer than the interval between successive peaks of the rectification waveform: $RC \gg T$; $RC \gg \frac{1}{f}$ [4]. Secondly, consider the ripple to achieve a better

DC conversion. The ripple voltage is $V_{r(p-p)} = \frac{I_{dc}}{fC}$. And ripple factor is basically a

measurement of ripple which denotes the purity of rectified output. Thus every effort is made to reduce the ripple factor. The ripple factor of half wave rectifier is $r = \frac{V_{rms}}{V_{dc}} =$

$$\frac{1}{RCf2\sqrt{3}} = 1.21. \text{ Combined with daily experience, if there is no delay device, the light will}$$

become flashing because of the intermittent irregular sound, and cannot achieve a good lighting effect. I am using this RC circuit to delay the lighting time by its charging and discharging character. Time constant of a parallel RC circuit is $\tau = R * C$; a whole period is about 5τ .

The load resistor is chosen to be 20kΩ. By using the overall resonating frequency at 540Hz and through trial and detect, the capacitor with the capacitance that satisfy the requirement is 100µF as $RC = 2 \text{ s}$ is much greater than ($\frac{1}{f}$ is

$$\frac{1}{540}) 0.00185 \text{ s}. \text{ The transfer function of this part is } H(s) = \frac{R_5}{1 + R_5 C s}.$$

For the diode of the rectifier, the 1N4148 diode was chosen due to fast switch speed and high conductance.

Non-inverting voltage Comparator

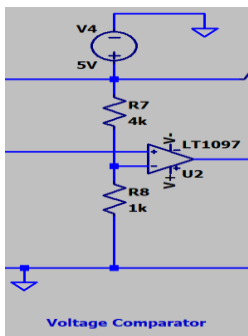


Figure b 5

In a non-inverting comparator, the reference voltage is applied to the inverting input and the voltage to be compared to the non-inverting input. When the voltage to be compared (V_{in}) exceeds the reference voltage, the output of the op-amp oscillates to positive saturation (V_+) and vice versa. Sine waves can be changed into square or rectangular waves of the same frequency.

As measured, the voltage entering the voltage comparator is around 3.2 volts, so the voltage of R_8 must be below 3.2 volts in order for the comparator to output high levels. Since R_7 and R_8 share a 5V source voltage, the voltage value ratio is equal to the ratio of the resistances of the two resistors. For convenience, set the R_7 to 4k ohms and the R_8 to 1k ohms, so that the R_8 only gets about 1 volt.

light control circuit

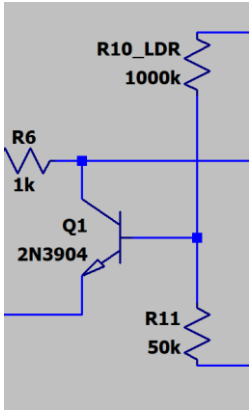


Figure b 6

R12 and photoresistor (LDR) are divided 5V. According to the characteristics of LDR, when the light intensity is strong in the daytime, the resistance of LDR is very small. Once the voltage at R12, the base of Q2, is higher than 0.7 volts, Q2 is forward active, making it impossible for any circuit behind Q2 to work. So that no matter what sound in the input, the LED light will never turn on.

After the sun sets and the light darkens, the resistance of the LDR reaches about 1million ohm, and the voltage divided by Q2 becomes very small, so that the circuit behind turns on.

led work part

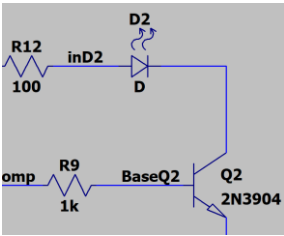


Figure b 7

This is the circuit drives the LED. Make sure that V_{inD2} and V_{BaseQ2} larger than 0.7 volte.

Power consumption calculated

Individual power consumption (mW)							
R1	0.66	R7	6.25	C1	0.65	Q2	33
R2	0.156	R8	1	C2	0	D1	1.92
R3	1.936	R9	10.64	C3	0.33	D2	34.2
R4	0.023	R10	0.024	U1	4.5		
R5	0.512	R11	1.13	U2	3.6		
R6	0.088	R12	210	Q1	0		

Simulation Results

Now I am going to check the performance of circuits by using LTspice .Because the circuit prioritizes only frequency, the linear relationship between output and input is not important for most parts of the circuit

RC Bandpass filter

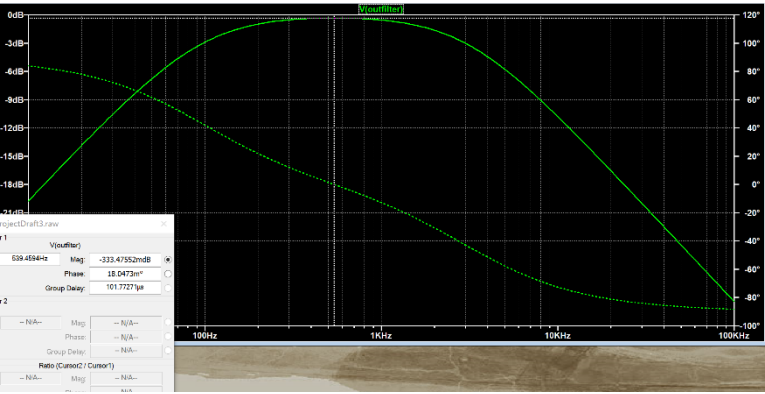


Figure c 1 :bode plot of the output in frequency domain

In figure c 1, we can see that the passband of this filter is from 100 Hz to 3k Hz with the center frequency 540 Hz. And it looks perfect.

To observe the linearity of the transfer characteristic of this filter, I use a sine wave signal with 540 Hz and a sine wave signal with 10k Hz as inputs.

When the input is at the same resonant frequency as the filter, the output voltage shows the expected linear behavior over the input voltage, though not perfectly.

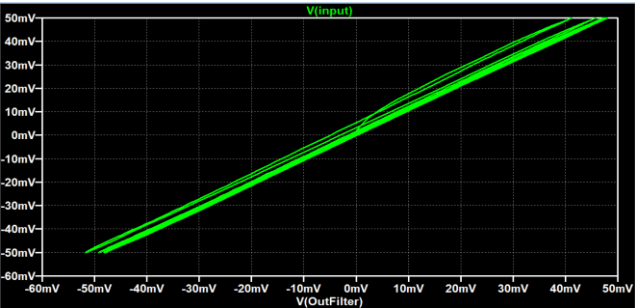


Figure c 2: Transfer characteristic of 540 Hz signal

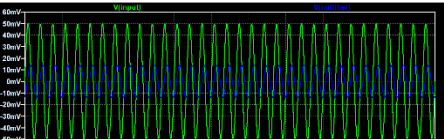


Figure c 4:in time domain

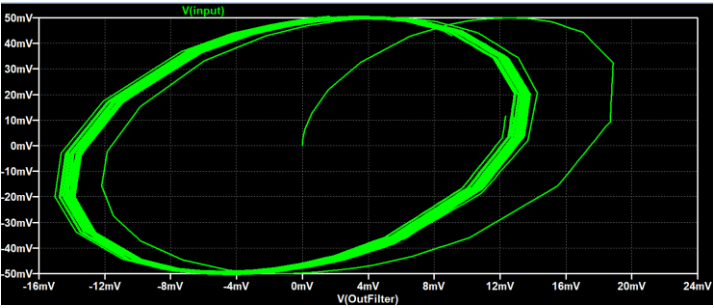


Figure c 3: Transfer characteristic of 10k Hz signal

However, when the frequency of the input voltage exceeds the bandwidth of the bandpass filter, the transmitted signal will be distorted and attenuated. And in time domain, the distortion and attenuation is also very obvious for input voltage which exceeds the bandwidth.

Non-inverting Operational Amplifier

To observe the gain of this part, I use a sine wave signal with 0.05v amplitude and 540 Hz. The expected magnification is about 86 times, and the signal will around 4 volts.

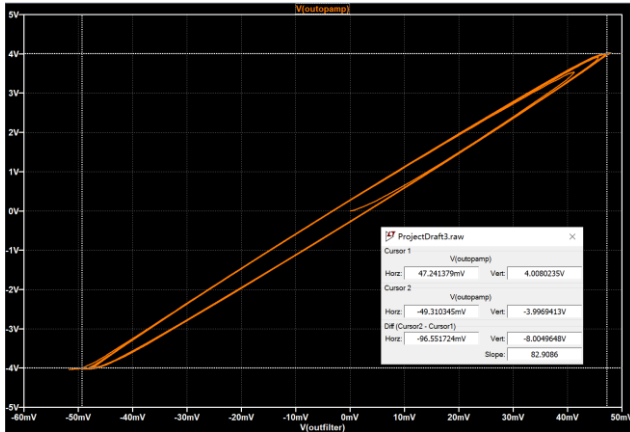


Figure c 5:voltage gain

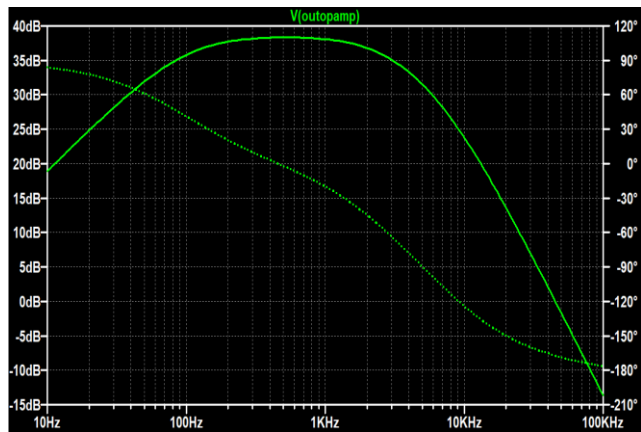


Figure c 6: bode plot of the output in frequency domain

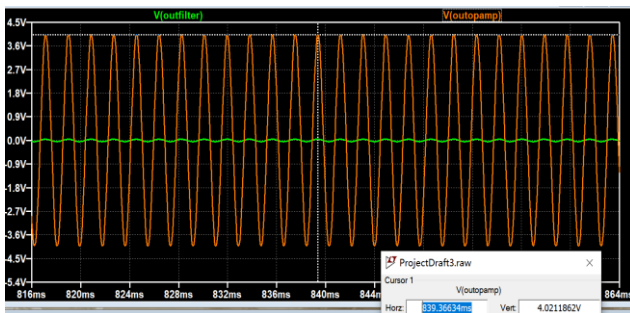


Figure c 7:input and output voltage in time domain

Half-wave rectifier and time-delay device

In figure c 8, according to the transmission characteristics of the rectifier, we can see that the AC signal becomes the DC signal. In figure c 9, a bit of a "jagged" shape to the output DC signal is to be expected, as mentioned above. In figure c 10, the fft of output voltage, the curve doesn't have a lot of "shake" and is more like a straight line because now it is DC signal. In figure c 11, the transmission characteristics are nonlinear is as expected because of charging and discharging of RC circuit.

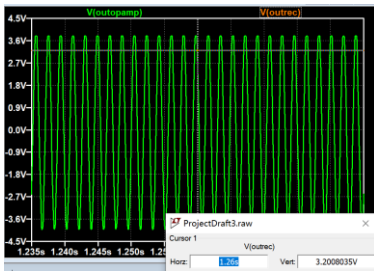


Figure c 8: AC to DC

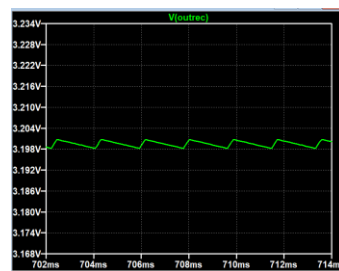


Figure c 9: output signal



Figure c 10:FFT of output

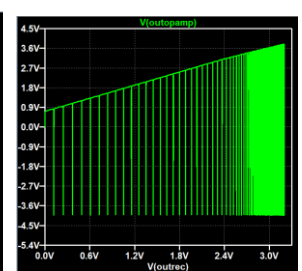


Figure c 11: Transfer characteristic

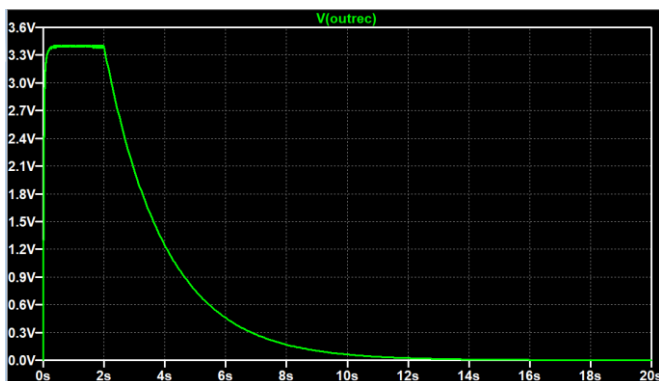


Figure c 12: check the time delay

I use transient Analysis to research the time delay of RC circuit. As can be seen from the figure c 12, its charging time is very short, and its discharge time is relatively long, about 10 seconds, which satisfies the design demands.

Non-inverting voltage Comparator

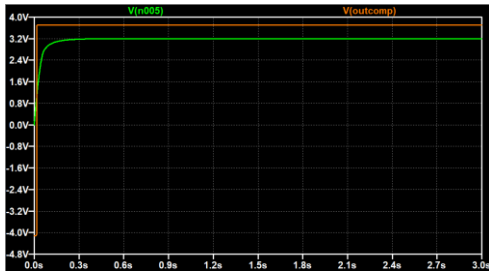


Figure c 13: change the shape

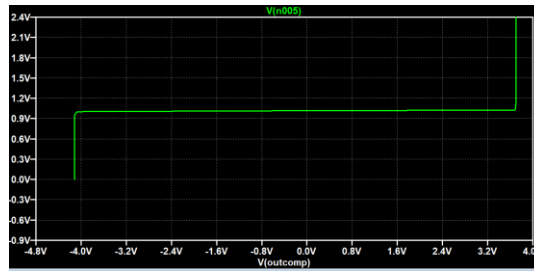


Figure c 14: Transfer characteristic

As shown in figure c 13, the waveform shape has been changed.

The transfer characteristic for input and output voltages is linear in figure c 14.

light control circuit

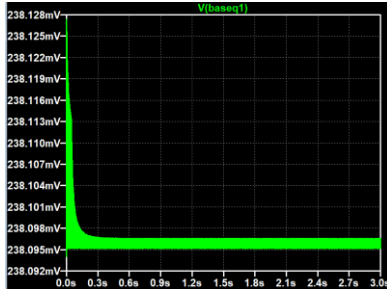


Figure c 15 : voltage of Q1 base

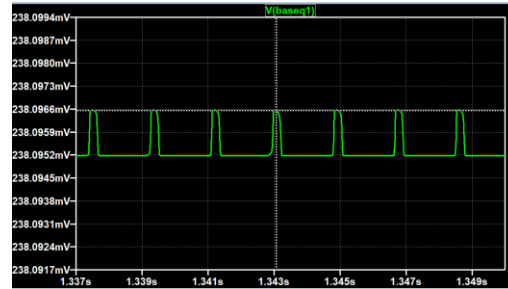


Figure c 16: voltage of Q1 base

The simulate situation here is at night when the resistance value of the photoresistor is very high, I assume it is 1000k ohms. The voltage at the base of Q1 is only 238 mV which is much less than 0.7 volts, so Q1 at cutoff state. The circuits behind can work.

The circuit that drives the led to work

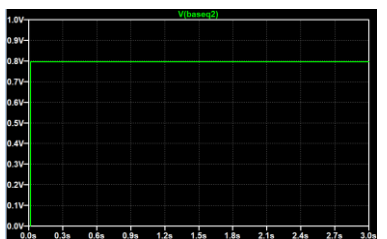


Figure c 17

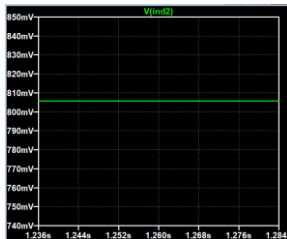


Figure c 18

The value of V_{ind2} and V_{BaseQ2} are larger than 0.7 volte. The NPN and led diode can work very well.

In summary, the circuit design is successful and supplies the requirements of use.

Power consumption measured

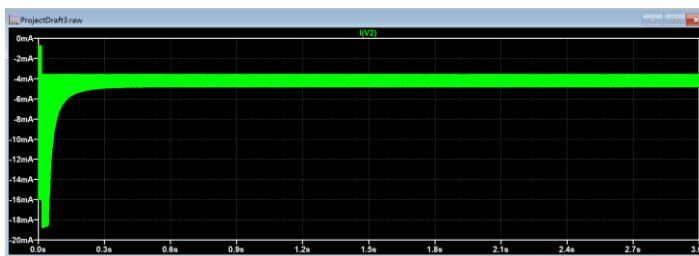


Figure d 1



Figure d 2

The current plot of the 5V supply which average current= - 4.2mA. Power dissipation of 5V source = $5V * -4.2mA = -21mW$. The current plot of the -5V supply which average current=0.715mA. Power dissipation of -5V source = $-5V * 0.715mA = -3.575mW$. Thus, the total power dissipation of this circuit is around -24.575mW

Reference

- [1]Kirt Blattenberger RF Cafe, "Standard Capacitor Values & Color Codes." [Online]. Available: <https://www.rfcafe.com/references/electrical/capacitor-values.htm>
- [2] Analog.com, 2020. [Online]. Available: <https://www.analog.com/media/en/technical-documentation/data-sheets/1097fas.pdf>. [Accessed: 05- Jun- 2020]
- [3]E. Tutorials, "Non-inverting Operational Amplifier - The Non-inverting Op-amp," *Basic Electronics Tutorials*, Feb. 2019. https://www.electronics-tutorials.ws/opamp/opamp_3.html
- [4]A. Electronics, "Half wave Rectifier Explained," *www.youtube.com*, Jul. 12, 2018. https://www.youtube.com/watch?v=LI0IOk_Ltfc&t=735s (accessed May 23, 2022).